PUBLIC PROJECT OVERVIEW

CV Optimizer

A practical job-search workspace for tailoring your CV to a role, preparing

for interviews, and keeping applications organized.

CV Optimizer combines an AI-assisted CV workflow with an application tracker. It is designed to help people prepare clearer, role-specific application materials while keeping the candidate in control of the final content.

> *AI output is a draft, not a hiring decision or a promise of an interview.*
>
> *Review every statement and make sure it accurately reflects your experience*
>
> *before sharing it with an employer.*

What you can do

- Analyze a PDF or DOCX CV against a job description and review potential matches, gaps, and suggestions.
- Generate a role-tailored CV, edit its contents, then download it as PDF or DOCX.
- Draft a company-specific cover letter, edit it, and download a PDF.
- Practice Indonesian- or English-language mock interviews with spoken questions, microphone answers, conversational AI follow-ups, and feedback.
- Track applications through stages such as applied, interview, offer, rejected, no response, and withdrawn.
- Browse up to 100 skill-ranked job listings for Indonesia, Malaysia, Singapore, Australia, and eligible remote roles. Visa sponsorship is flagged only when the listing text explicitly mentions it; review tailored materials and submit each application through the employer's official site.
- Select several listings and prepare separate tailored CVs plus cover letters in one sequential batch, capped by the account's remaining AI task quota.
- Use the optional local Chrome/Edge helper for supported Greenhouse and Lever forms to fill recognizable contact and cover-letter fields; review and submit each application yourself.

Some capabilities depend on an authenticated account and a configured AI provider. Recruiter sourcing does not currently provide a real candidate database; do not treat its empty results as a search of the wider labor market. Voice practice uses browser speech recognition and speech synthesis when available. Browser/device providers may process microphone audio to produce a transcript; the application sends the transcript to Gemini for feedback. This is turn-based practice, not live streaming or proctoring.

Product preview

The interface is primarily Indonesian. English-language support is partial. Screenshots can be added here as the public demo and its test data are finalized.

Project highlights

- React, TypeScript, and Vite frontend with responsive, route-based pages.
- Express and TypeScript API, PostgreSQL persistence, and Prisma migrations.
- Redis and BullMQ for asynchronous AI tasks, processed by a separate worker.
- AI provider abstraction with Google Gemini integration.
- Google sign-in, server-side monthly AI quota reservations, rate limiting,

file validation, and typed API errors.

- Docker Compose for running the full local stack.
- Automated backend tests and frontend lint/build checks in GitHub Actions.

Run locally

Requirements

- Docker Desktop with Docker Compose
- A Google Gemini API key

Start the application

```
Copy-Item .env.example .env
```

Edit .env and provide GEMINI_API_KEY, JWT_SECRET, and AI_JOB_ENCRYPTION_KEY. Generate a JWT secret and a separate encryption key:

```
node -e "console.log(require('crypto').randomBytes(48).toString('base64url'))"
node -e "console.log(require('crypto').randomBytes(32).toString('base64'))"
```

Use the first value for JWT_SECRET and the second for AI_JOB_ENCRYPTION_KEY. Keep both private. Then start the services:

```
docker compose up --build
```

- Web app: <http://localhost:5173>
- API health: <http://localhost:4000/api/health>
- API readiness: <http://localhost:4000/api/ready>

Google sign-in also needs a Google OAuth Web Client ID. Add the frontend origin to the OAuth client's authorized JavaScript origins, then set GOOGLE_CLIENT_ID in .env. For local use, the frontend origin is http://localhost:5173.

To stop the stack without deleting database data:

```
docker compose down
```

Avoid docker compose down -v unless you intentionally want to delete local database and Redis volumes.

For manual setup, environment reference, API contracts, and deployment instructions, see the technical guide.

Documentation

- Technical guide (Indonesian) - architecture,
data flow, API map, setup, operations, and a learning path.

- Project overview PDF - public
portfolio-friendly summary.

- Technical guide PDF - printable technical
companion to the guide.

PDFs can be regenerated after editing either Markdown document:

```
cd backend
npm run docs:pdf
```

Quality checks

```
cd backend
npm ci
npm test

cd ../frontend
npm ci
npm run check
```

Important limitations

- AI analysis and generated text can be incorrect. Verify facts, dates,
skills, and claims before using an output.

- A match score is an estimate based on the provided CV and job description;
it is not a probability of being hired.

- AI tasks require a working Gemini configuration and are subject to provider
availability, quotas, costs, and policies.

- International country listings use the optional Jooble API key
(JOOBLE_API_KEY); without it, country coverage outside Indonesia is limited to eligible worldwide remote listings. Job-board coverage is not exhaustive, visa sponsorship flags are text hints rather than employer verification, and applications are not automatically submitted. The optional browser helper supports only a small set of Greenhouse/Lever fields and does not upload CV files, answer screening questions, or bypass CAPTCHA. See browser-extension/README.md for local installation.

- Recruiter candidate sourcing is not connected to a consent-based candidate database.
- The local Docker Compose setup is for development and demonstration. It is not, by itself, a production deployment or evidence of capacity for a particular number of users.
- Before a public production launch, configure OAuth for the real domain, review privacy/legal disclosures, deploy managed database and Redis services, and run end-to-end and load tests against the chosen hosting tier.

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